GRASP4Figs: OCR-Augmented Reward-Conditioned Captioning for Scientific Figures

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ABSTRACT

In this work, we present **GRASP4Figs**, two case studies in scientific figure captioning unified by a shared diagnostic methodology: identify why caption generation fails at a given output length, then apply the remedy that failure mode specifically demands. Generating faithful, informative textual descriptions of scientific figures is a formidable challenge in multimodal document understanding, as figures encode high-density information through axis labels, legend entries, and mathematical notation that general-purpose vision-language models are systematically ill-equipped to interpret. While reward-conditioned frameworks such as FigCaps-HF supervise caption quality using human feedback, they process figures without access to this embedded textual content and remain restricted to short, superficial captions. To address these limitations, we propose GRASP4Figs to inject OCR-derived context directly into the captioning pipeline and extend generation to paragraph-length analytical description. In **Case 1**, we augment the BLIP vision-language model with a novel *OCR-context injection* mechanism that prepends tokenized in-figure text to the training prompt, combined with a composite reward signal within an offline Upside-Down Reinforcement Learning from Human Feedback (UDRL-HF) framework. In **Case 2**, we identify a critical failure mode, mention-paragraph-induced grounding collapse (MPGC), where fine-tuning Qwen2.5-VL-7B on SciCap+ mention-paragraphs induces hallucination of unobservable experimental methodology, and instead adopt zero-shot inference with structured visual-grounding prompts. Additionally, we develop an end-to-end web application, built on the Case 2 zero-shot pipeline, with a per-figure interactive question-answering chatbot. Experimental results on the complete FigCaps-HF benchmark of 13,355 scientific figures demonstrate improved performance for GRASP4Figs over baselines, yielding a BLEU-4 of **0.0373** (**166.4%** over FigCaps-HF's published non-RLHF fine-tuning baseline and **96.3%** over its strongest RLHF baseline, with gains of +23.3% in BLEU-1 and +39.7% in METEOR) in Case 1, and a ROUGE-L Recall of **0.3782** with analytically rich, 176-word paragraphs in Case 2.

CCS CONCEPTS

• **Computing methodologies** → **Computer vision**; **Natural language generation**; *Reinforcement learning*; *Neural networks*; • **Applied computing** → *Optical character recognition.*

ICVGIP2026, December 21-25, 2026, Kolkata, India

ACM ISBN 978-x-xxxx-xxxx-x/YY/MM.
https://doi.org/10.1145/nnnnnnn.nnnnnnn

KEYWORDS

scientific figure captioning, vision-language models, OCR augmentation, UDRL-HF, reward-conditioned generation, zero-shot captioning, hallucination, Qwen2.5-VL, FigCaps-HF

ACM Reference Format:
Submission Id XYZW. 2026. GRASP4Figs: OCR-Augmented Reward-Conditioned Captioning for Scientific Figures. In *Proceedings of 17th Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP2026).* ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 INTRODUCTION

In this work, we present GRASP4Figs, two case studies in scientific figure captioning, at short-caption and paragraph length, unified by one diagnostic methodology: identify why generation fails at that output length, then apply the remedy it demands. Scientific figures, quantitative plots, architecture diagrams, performance curves, and ablation charts, are the primary medium through which researchers communicate empirical evidence [5, 13], yet automated generation of faithful, informative descriptions of such figures remains substantially unsolved: figures encode high-density information through axis labels, mathematical notation, and color-coded legends that general-purpose vision-language models (VLMs), trained on web-scale corpora, are systematically ill-equipped to interpret [17, 22].

While reward-conditioned frameworks such as FigCaps-HF [28] exist, most are restricted to BLIP's 384×384 image encoding, which cannot reliably resolve small-font axis labels, so they generate captions describing visual gestalt (e.g., *'a graph with lines"*) rather than semantic content (e.g., *'outage probability decreasing as SNR increases from −40 to +40 dB"*) [13]. Existing work also targets only short, 10–50 word captions, leaving 100–200 word analytical explanations unaddressed; the natural training source for such outputs, SciCap+ mention-paragraphs [30], has not been critically evaluated and, as we show, is actively harmful, compounding known hallucination behavior in VLMs [14, 27].

To address these limitations, we observe that the bottleneck lies not in the language decoder's generative capacity but in the vision encoder's inability to resolve fine-grained textual content at low resolution. Rather than modifying BLIP's architecture or upsampling its visual input, we instead supply this missing textual evidence externally, letting the decoder condition on axis labels, legend entries, and in-figure annotations the vision encoder cannot recover: a novel OCR-context injection mechanism that prepends tokenized in-figure OCR text to the training prompt, requiring no architectural change and no external OCR at inference. Because faithfulness alone does not guarantee informativeness, we further combine this with a composite UDRL-HF reward that jointly optimizes for human-rated helpfulness and caption length.

Extending this framework to paragraph-length description, however, introduces a distinct failure mode absent from short-caption generation. Naively fine-tuning on SciCap+ mention-paragraphs, the most direct source of paragraph-length supervision, induces the model to hallucinate unobservable experimental methodology, since mention-paragraphs describe surrounding context, dataset details and training procedures, not visually grounded in the figure itself. We term this failure mode mention-paragraph-induced grounding collapse (MPGC). Motivated by this finding, we instead adopt zero-shot inference with structured visual-grounding prompts, which constrain generation to only visually verifiable content and avoid this hallucination entirely. Specifically, the main contributions of our work are:

- **Problem and Case Studies.** We present **GRASP4Figs**, two case studies unified by a shared diagnostic methodology: unresolved in-figure text at low resolution (short captions), and the lack of paragraph-length, grounded description (long-form). GRASP4Figs contributes:
  - **OCR-Context Injection**, a data augmentation method that prepends tokenized in-figure OCR text to BLIP's training prompts, requiring no architectural modification and no external OCR at inference [6, 7].
  - **Composite UDRL Reward Design**, combining human feedback helpfulness scores with a detail-aware length reward to steer generation toward more informative descriptions.
  - **MPGC Characterization and Zero-Shot Paragraph Generation**, the first formal documentation of mention-paragraph-induced grounding collapse, motivating a zero-shot, structured visual-grounding prompting strategy for Qwen2.5-VL-7B [26] that avoids it.
- **Results.** On the full 13,355-figure FigCaps-HF benchmark, GRASP4Figs achieves a BLEU-4 of 0.0373 (166.4% over the non-RLHF baseline, 96.3% over the strongest RLHF baseline) and a ROUGE-L Recall of 0.3782 on 176-word analytical paragraphs.

2 RELATED WORK

In this section, we discuss recent works on scientific figure captioning and multimodal document understanding. We broadly classify related work into dataset and benchmark construction, document and chart understanding models, general-purpose vision-language models, and reward-conditioned generation with hallucination mitigation.

Dataset and Benchmark Construction. SciCap [5] introduced the first large-scale figure-caption dataset from arXiv papers, establishing CNN+LSTM baselines showing that textual context substantially outperforms vision-only approaches. FigCaps-HF [28] further advanced this direction by proposing UDRL-HF, training an auxiliary BERT-based scorer on expert annotations to infer reward labels for the full corpus, and conditioning caption generation on control tokens; their best result achieves BLEU-4 of 0.019 and ROUGE-L of 0.152. SciCap+ [30] augments this benchmark with mention-paragraphs and pre-extracted OCR tokens, while Huang et al. [9] showed that approximately 75% of caption words appear verbatim in mention-paragraphs, highlighting the importance of textual grounding in caption generation. However, these benchmarks target only short, 10–50 word captions, leaving longer analytical description largely unstudied.

Document and Chart Understanding Models. Chart- and document-specific pretraining has produced a range of specialized architectures: MatCha [17] and UniChart [22] target mathematical and structural chart reasoning; Pix2Struct [11] and Donut [10] parse rendered visual input directly, the latter eliminating OCR entirely; DePlot [18] and ChartQA [23] focus on plot-to-table translation and logical reasoning; and DocOwl [6, 7] incorporates textual context for document understanding. Despite this progress, these models are primarily evaluated on structured question answering or table extraction rather than free-form analytical description.

General-Purpose Vision-Language Models. Large-scale vision-language pretraining has produced increasingly capable captioning backbones, including BLIP [13], BLIP-2 [12], Flamingo [1], and instruction-tuned models such as LLaVA [19]. These models generalize well to natural images, but their vision encoders, trained on web-scale photographic corpora, are systematically ill-equipped to resolve the small-font axis labels, legend entries, and symbolic notation embedded in scientific figures, motivating figure-specific augmentation rather than generic pretraining alone.

Reward-Conditioned Generation and Hallucination. Beyond captioning, reward-conditioned generation has been widely explored for aligning model outputs with human preference, most notably through reinforcement learning from human feedback [24] and its offline, upside-down formulation [29] adopted by FigCaps-HF. Separately, hallucination remains a well-documented failure mode in both image captioning [27] and large vision-language models more broadly [15, 20]. Methods such as Chain-of-Region verification [14] attempt to enforce grounding by cross-checking generated text against localized visual regions, yet faithful alignment remains difficult for long-form generation, where hallucinated content is harder to localize and penalize.

Despite these advancements, existing approaches exhibit several limitations. The dominant paradigm relies on models that do not effectively utilize the rich textual information embedded within figures, leading to superficial caption generation, while most works remain restricted to short captions or structured question answering rather than analytical, paragraph-length description. In this work, we propose GRASP4Figs, a framework that injects OCR-derived context directly into reward-conditioned caption generation and characterizes, for the first time, how naturally-occurring long-form training text induces hallucination in scientific figure description.

3 METHODOLOGY

In this section, we present the methodology of the proposed GRASP4Figs framework, describing the overall approach, key techniques, and implementation details used for scientific figure captioning and paragraph generation. As motivated in Section 1, we present GRASP4Figs as two case studies rather than a single end-to-end solution: Case 1 studies short-caption generation, augmenting a compact captioning backbone with OCR-derived context and a composite reward signal, while Case 2 studies paragraph-length generation, introducing zero-shot structured prompting to avoid a previously undocumented hallucination failure mode. Figure 1 illustrates the resulting Case 1 core architecture: three input streams, visual embeddings from ViT-Base [4], OCR context tokens extracted from figure annotations, and a UDRL composite reward control token, are fused and jointly processed by the BLIP-Base decoder to generate captions conditioned on visual content, OCR grounding, and human feedback signals.

This case-study structure follows directly from an empirical observation: short captions and long-form paragraphs fail for different reasons, and therefore demand different remedies. Short captions fail primarily because the vision encoder cannot resolve fine-grained textual evidence, axis labels, legend entries, tick values, a problem of *missing evidence* that reward-conditioned fine-tuning with explicit OCR grounding is well-suited to correct, since model capacity is adequate and only the input signal is incomplete. Paragraph-length descriptions fail for a qualitatively different reason: fine-tuning directly on mention-paragraph text teaches the model to imitate discourse structure only loosely coupled to the image, a problem of *misaligned training signal* that no amount of reward shaping or OCR grounding can correct, since the defect lies in what the model is taught to imitate rather than what evidence it can access. Case 1 and Case 2 therefore pursue the same goal, faithful, information-rich description, with remedies matched to each regime's specific failure mode.

3.1 Case 1: OCR-Augmented UDRL-HF Caption Generation

In this subsection, we describe how OCR-derived textual context and a composite reward signal are combined to steer BLIP's caption generation toward more informative, faithful descriptions. The resulting pipeline unfolds in three stages: offline preprocessing, comprising OCR extraction, composite reward computation, quality filtering, and UDRL token assignment; UDRL-HF training on the resulting OCR-augmented inputs; and inference on the full 13,355-figure test set. Algorithm 1 below formalizes the first two stages precisely; Section 4.1 reports the third.

Two design choices distinguish this pipeline from a direct application of UDRL-HF to FigCaps-HF, each targeting a failure mode the other cannot reach. First, the benchmark's helpfulness scorer rewards semantic alignment with human judgments but cannot penalize captions that are accurate yet uninformatively short or repetitive; we therefore compose it with a length-aware reward, using the resulting composite score in two distinct roles, relabeling the UDRL control token at one threshold and filtering low-quality pairs at a stricter, independent threshold, so labeling and data-cleaning remain separable rather than conflated. Second, because the helpfulness scorer operates purely on caption text, it cannot correct a caption that is fluent but factually detached from the figure, the resolution bottleneck motivated in Section 1; OCR-context injection addresses this complementary axis by supplying the decoder with textual evidence the vision encoder cannot resolve. The two mechanisms therefore address non-overlapping deficiencies, informativeness/length versus factual grounding, and Section 4.2 verifies that their combination, not either alone, accounts for the reported gain.

3.1.1 *OCR-Context Injection.*

To supply the language decoder with textual evidence that the vision encoder alone cannot resolve, we make use of the `Img-text` field already provided by the FigCaps-HF benchmark for each figure. This field contains tokenized strings extracted by OCR, axis label text, numeric tick values, legend entries, and title strings, yet it remains entirely unused in the baseline system. Rather than modifying BLIP's vision encoder to resolve this information visually, we instead inject it directly as text: we prepend a selection of these tokens to the training input, constructing:

$$\text{input} = \underbrace{\texttt{Figure text: } t_1\, t_2\, \ldots\, t_k}_{\text{OCR context } (k=25)} \;.\; \underbrace{\texttt{[GOOD/BAD]}}_{\text{reward token}}\; \underbrace{c_1\, c_2\, \ldots\, c_n}_{\text{caption tokens}} \tag{1}$$

We limit this context to k=25 tokens per figure, which prevents OCR content from dominating BLIP's attention budget while keeping the total input within the 512-token maximum sequence length. Crucially, no external OCR engine is required at inference: the `Img-text` annotations are read directly from the benchmark JSON, so the mechanism adds no additional dependency to the deployed pipeline. This design directly addresses the fundamental limitation motivated in Section 1, BLIP's 384×384 resolution encoding cannot reliably resolve small-font axis labels, by circumventing the vision encoder's resolution bottleneck entirely rather than attempting to fix it.

3.1.2 *Composite Reward Signal.*

Injecting OCR context alone addresses only faithfulness; it does not by itself guarantee that the resulting captions are informative. To steer generation toward more informative descriptions, we build on the reward-conditioning mechanism already used by FigCaps-HF, which trains a BERT-based [3] auxiliary regressor on 439 expert-annotated pairs to predict a helpfulness score $h \in [0, 1]$ per caption. We augment this helpfulness signal with a detail-aware length reward that explicitly favors captions of moderate, information-dense length over both terse and overly verbose outputs:

$$\lambda(c) = \begin{cases} 0.1 & |c| < 10 \\ 0.3 + 0.04(|c| - 10) & 10 \le |c| < 20 \\ 0.7 + 0.01(|c| - 20) & 20 \le |c| < 40 \\ 1.0 & 40 \le |c| \le 60 \\ \max\big(0.4,\ 1.0 - 0.012(|c| - 60)\big) & \text{otherwise} \end{cases} \tag{2}$$

where $|c|$ is the caption word count and the reward peaks over a 40–60 word plateau, matching the short-caption regime targeted by Case 1. The two signals are combined into a single composite reward, computed once for the entire dataset as an offline preprocessing pass:

$$r(c) = (1 - \alpha) \cdot h(c) + \alpha \cdot \lambda(c), \qquad \alpha = 0.3 \tag{3}$$

with $\alpha = 0.3$ chosen to let helpfulness dominate while still meaningfully rewarding descriptive length. We use this composite score for two distinct purposes, governed by two distinct thresholds. First, we relabel each pair's `[GOOD]/[BAD]` control token by thresholding the composite score at $\beta = 0.55$, replacing FigCaps-HF's original helpfulness-only labeling with one that also reflects caption length. Second, at training time we apply a stricter quality filter, discarding pairs with $r(c) < \tau = 0.35$ entirely rather than merely relabeling them, eliminating 3,029 of 106,834 training

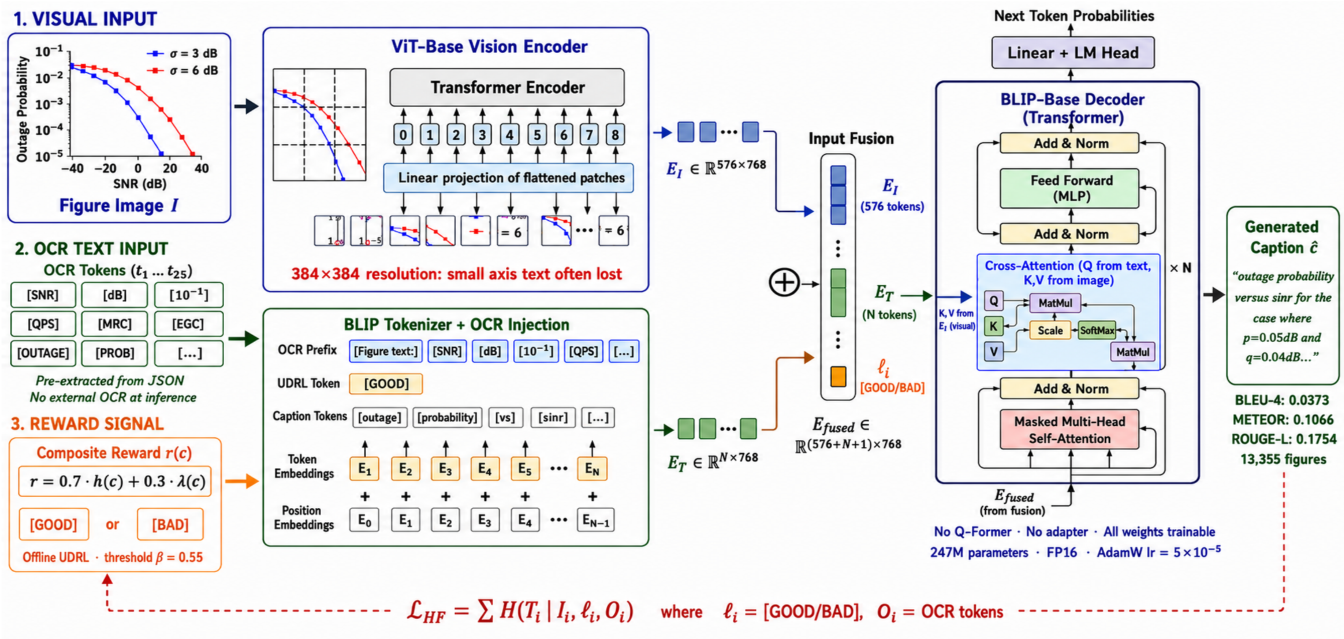

Figure 1: GRASP4Figs Case 1 core architecture. Three input streams visual embeddings from ViT-Base, OCR context tokens from figure annotations, and a UDRL composite reward control token are fused and processed by the BLIP-Base decoder to generate captions conditioned on visual content, OCR grounding, and human feedback signals. No Q-Former, no adapter; all weights are trainable. Case 2 (Section 3.2) uses a separate, frozen vision-language model and is not depicted here.

pairs (2.8%) that are neither helpful nor sufficiently descriptive. Because $\tau < \beta$, pairs scoring between the two thresholds survive filtering while retaining a `[BAD]` label, so the training set preserves both control-token classes rather than collapsing every retained example to `[GOOD]`. Algorithm 1 details the resulting two-stage procedure: an offline composite-relabeling pass over the full dataset, followed by OCR-augmented fine-tuning on the filtered result.

3.1.3 Training Configuration. Having established the OCR-injection and composite reward mechanisms, we now detail the configuration used to fine-tune Case 1. We fine-tune `Salesforce/blip-image-captioning-base` (250M parameters) with HuggingFace Accelerate, FP16 mixed precision, AdamW optimizer [21] (lr = 5×10^{-5}), batch size 2, maximum sequence length 512 tokens, for 10 epochs on an NVIDIA RTX 6000 Ada (48 GB VRAM). All metrics use `rouge-score` v0.1.2 and NLTK `corpus_bleu` with Method1 smoothing, identical to the original FigCaps-HF evaluation stack, to ensure our reported gains are directly comparable to published baselines.

Algorithm 1 OCR-Augmented Composite-Reward Training (Case 1)

Require: FigCaps-HF pairs $\mathcal{D} = \{(I_i, T_i, \mathrm{OCR}_i)\}$, BERT scorer $\mathcal{R}$, label threshold $\beta = 0.55$, filter threshold $\tau = 0.35$, $\alpha = 0.3$, $k = 25$ OCR tokens
Ensure: Trained BLIP checkpoint θ^*

```
// Stage 1: Offline Composite Relabeling (run once)
for each pair (I_i, T_i) ∈ D do
    h_i ← R(T_i)                                   ▷ BERT helpfulness score
    λ_i ← λ(T_i)                                   ▷ Length reward (Eq. 2)
    r_i ← (1 − α) · h_i + α · λ_i                  ▷ Composite reward (Eq. 3)
    ℓ_i ← [GOOD] if r_i ≥ β else [BAD]             ▷ Relabel control token
    Write (r_i, ℓ_i) back to pair i's record
end for
// Stage 2: OCR-Augmented Fine-tuning
D′ ← {i ∈ D : r_i ≥ τ}                             ▷ Quality filter; |D′| = 103,805
Initialize θ ← blip-image-captioning-base
for epoch = 1 to 10 do
    for each pair i ∈ D′ do
        o_i ← (t_1, t_2, ..., t_k) from OCR_i      ▷ First k tokens
        x_i ← Figure text: o_i. ℓ_i T_i            ▷ Construct input (Eq. 1), truncate to 512 tokens
        T̂_i ← BLIP_θ(I_i, x_i)                     ▷ Image-conditioned decoder forward pass
        L ← CrossEntropy(T̂_i, x_i)                 ▷ Teacher-forced captioning loss
        Update θ via AdamW (lr = 5 × 10^−5)
    end for
end for
return θ* ← θ
```

3.2 Case 2: Zero-Shot Paragraph Generation

Having established Case 1's short-caption improvements, we next turn to the substantially harder problem of paragraph-length analytical description, which, as we show below, surfaces a failure mode entirely absent from short-caption generation. Figure 2 illustrates the resulting Case 2 core architecture, arrived at after the fine-tuning failure diagnosed below.

We do not extend the Case 1 BLIP backbone to this setting. BLIP-Base's decoder is trained end-to-end on short, single-sentence targets, and its 384×384 vision encoder, already identified in Section 3.1 as a resolution bottleneck even for short captions, offers no additional benefit for the qualitatively different demands of paragraph-length description, coherent multi-sentence

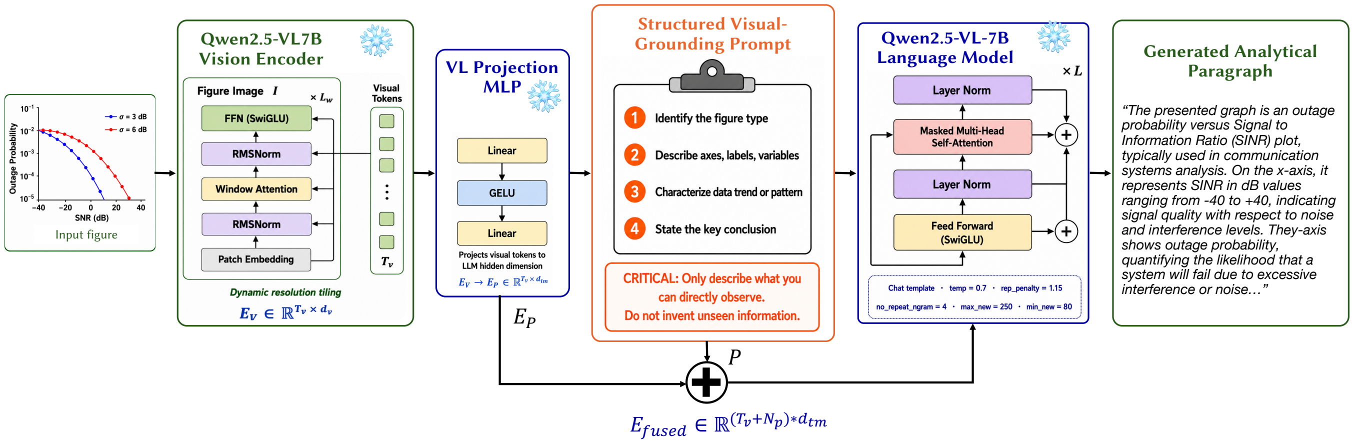

Figure 2: GRASP4Figs Case 2 core architecture. The frozen Qwen2.5-VL-7B-Instruct vision-language model is constrained entirely through a structured visual-grounding prompt, with no gradient update and no fine-tuning at any stage. Case 1 (Figure 1) uses a separate, trainable BLIP-based architecture and is not depicted here.

discourse, trend articulation spanning the full figure, and conclusion synthesis, capabilities a large-scale, instruction-tuned vision-language model already possesses out of the box. We therefore adopt `Qwen2.5-VL-7B-Instruct` [26] as the Case 2 backbone and, following the same logic that motivated Case 1's design, first adapt it to the task with the most direct available recipe: supervised fine-tuning on paragraph-length ground truth. As we show next, this direct approach fails in a specific, diagnosable way, and it is that diagnosis, rather than an a priori architectural preference, that determines Case 2's final design.

3.2.1 Mention-Paragraph-Induced Grounding Collapse. Paragraph-length generation requires a paragraph-length supervision source, and SciCap+ [30] provides the most natural candidate: mention-paragraphs, excerpts from paper body text referencing each figure, typically 80–200 words in length. Following the standard fine-tuning recipe, we fine-tuned `Qwen2.5-VL-7B-Instruct` [26] on 50,000 such pairs using LoRA [8] (r=16, α=32, 4-bit NF4 quantization, lr=1×10^{-4}, 3 epochs).

Contrary to expectation, qualitative evaluation revealed catastrophic **MPGC** rather than the anticipated improvement in descriptive richness. For a figure whose ground truth is *"Fig. 7. Evolution of GP Function avgD"*, the model produced:

> *"The probes are initially placed uniformly on the x-axis. The probes are then moved by the same amount in the direction of the gradient. The probes are then sorted by their distance from the best probe. The probes are then moved again…"*

This describes experimental methodology nowhere visible in the figure, a clear symptom of the model reproducing the register of its training text rather than describing the image in front of it. We identify three causal factors underlying this collapse: (1) *distributional mismatch*, mention-paragraphs encode unobservable information from the paper body that no vision encoder could recover; (2) *linguistic prior dominance*, the high fluency of the training targets overwhelms the visual grounding objective during fine-tuning; and (3) *LoRA textual overfitting*, the adapters learn the writing register of methodology sections while effectively ignoring the image tokens.

3.2.2 Zero-Shot Structured Prompting. Having established that mention-paragraph fine-tuning is counter-productive rather than merely suboptimal, we abandon fine-tuning on this supervision signal entirely and instead evaluate zero-shot `Qwen2.5-VL-7B-Instruct`, constraining its output through a structured visual-grounding prompt rather than through gradient updates. The prompt decomposes generation into five ordered components: (1) figure type identification; (2) axis and variable description; (3) primary trend characterization; (4) conclusion articulation; and (5) a grounding constraint: *"Only describe what you can directly observe. Do not invent experimental details not shown in the image."*

Algorithm 2 formalizes the resulting inference procedure, in which all Qwen2.5-VL-7B weights remain frozen and generation is controlled entirely through the prompt structure rather than through any parameter update.

This design targets the causal factors identified above directly. Because the model's weights remain frozen, no gradient update occurs, so neither the linguistic-prior dominance nor the LoRA textual overfitting that produced MPGC can arise; there is simply no mechanism through which the fluency of a paragraph-length supervision signal could overwrite the model's pretrained visual grounding. The explicit grounding constraint in the prompt then substitutes, at inference time, for the task-specific supervision we deliberately withhold, instructing the model to rely on its own pretrained grounding rather than on paragraph-specific fine-tuning that we have shown, empirically, to be actively harmful.

We report ROUGE-L Recall as the primary Case 2 metric because ground-truth captions average 10–15 words while our outputs average 176.1 words; the precision component of F1 is systematically deflated by this length mismatch and does not reflect semantic quality.

Algorithm 2 Zero-Shot Paragraph Generation with MPGC Avoidance (Case 2)

Require: Test figure I, structured prompt template $\mathcal{P}$, frozen model `Qwen2.5-VL-7B-Instruct`
Ensure: Grounded analytical paragraph $\hat{G}$

```
// Construct structured prompt
P ← 𝒫.FORMAT({
    "Identify figure type,"
    "Describe axes and variables,"
    "Characterize primary trend,"
    "State key conclusion,"
    "Only describe what you can directly observe.
       Do not invent unseen information."})
// Vision encoding (all weights frozen)
E_V ← QwenViT(I) ∈ ℝ^{T_v×d_v}                ▷ Dynamic resolution tiling
E_P ← VLProjectionMLP(E_V) ∈ ℝ^{T_v×d_lm}     ▷ Frozen projection layer
// Prompt encoding
E_text ← Tokenizer(P)
E_fused ← [E_P ⊕ E_text]
// Constrained generation
Ĝ ← QwenLLM(E_fused,
      temp=0.7, rep_penalty=1.15, no_repeat_ngram=4,
      max_new_tokens=250, min_new_tokens=80)   ▷ Length floor enforced
   directly by the decoder, not a post-hoc check
return Ĝ
```

3.3 Experimental Setup

To ensure the reported results are reproducible and directly comparable to prior work, we detail below the hardware, software, dataset usage, and evaluation protocol used across both cases. All Case 1 training and evaluation were conducted on a single NVIDIA RTX 6000 Ada GPU (48 GB VRAM), with full test-set evaluation covering all 13,355 figures. Case 2 zero-shot inference using Qwen2.5-VL-7B-Instruct [26] was executed on a 3-GPU RTX 6000 Ada cluster (48 GB each) at FP16 precision.

The two datasets used in this work play deliberately distinct roles rather than being interchangeable sources of training data. FigCaps-HF [28] supplies both the Case 1 training pairs and the evaluation ground truth: every quantitative result reported in Section 4, for Case 1 and Case 2 alike, is scored against the same FigCaps-HF 13,355-figure test set, so all comparisons across both cases share one consistent reference. SciCap+ [30] mention-paragraphs, by contrast, are used exclusively as a training signal, specifically for the Case 2 fine-tuning experiment described in Section 3.2; we never evaluate against SciCap+ references directly, since doing so would conflate the MPGC failure mode with a dataset-mismatch artifact rather than isolating it as a property of the fine-tuning procedure itself.

Experiments were implemented using PyTorch 2.x with HuggingFace Transformers and Accelerate for training orchestration. Case 1 uses FP16 mixed precision with BLIP [13], while Case 2 employs Qwen2.5-VL-7B-Instruct [26] with BitsAndBytes 4-bit NF4 quantization during LoRA-based experiments [8]. Evaluation follows the FigCaps-HF protocol [28], using `rougescore` [16], NLTK `corpus_bleu` [25], and `meteor_score` [2] with standard tokenization and smoothing to ensure direct numerical comparability.

We deliberately do not force a single evaluation protocol across both cases: Case 1 reuses FigCaps-HF's exact metrics and tokenization for direct comparability to Singh et al. [28], while Case 2 reports ROUGE-L Recall for the length-mismatch reasons given in Section 3.2. Both choices follow the same principle, matching the protocol to the empirical question each case poses rather than fixing it for uniformity's sake.

4 RESULTS AND ANALYSIS

Section 3 argued that Case 1 and Case 2 pursue the same underlying goal, faithful, information-rich figure description, through two remedies matched to two distinct failure modes. We now evaluate whether each remedy actually delivers on that claim, in five steps. Section 4.1 tests whether Case 1's OCR-augmented composite-reward training improves short-caption quality over every published FigCaps-HF baseline; Section 4.2 isolates which of Case 1's two mechanisms, OCR injection or the composite reward, is responsible for that gain; Section 4.3 tests whether Case 2's zero-shot structured prompting succeeds where mention-paragraph fine-tuning collapsed into MPGC; and Sections 4.4 and 4.5 close with example-level and perception-level evidence, respectively, corroborating the aggregate metrics reported in the sections above them.

4.1 Case 1 Quantitative Results

Having described the OCR-injection and composite reward mechanisms in Section 3.1, we now evaluate whether they translate into measurable captioning improvements. Table 1 reports GRASP4Figs Case 1 against all published FigCaps-HF baselines on the full 13,355-figure test set, using the identical evaluation protocol described in Section 3.1 to ensure the comparison is direct.

Table 1: Case 1 quantitative results on FigCaps-HF test set (13,355 figures). Best result per column in bold. Δ denotes improvement over BLIP-RLHF Helpfulness (B-4, R-L) or, where Helpfulness reports no value, BLIP Fine-tune (B-1, MET), the only baseline supplying those columns.

Model	B-1	B-4	MET	R-1	R-2	R-L
BLIP Fine-tune [28]	0.2039	0.0140	0.0763	–	–	–
BLIP-RLHF Help. [28]	–	0.0190	–	–	–	0.1520
BLIP-RLHF Take. [28]	–	0.0230	–	–	–	0.1676
Ours (Case 1)	**0.2514**	**0.0373**	**0.1066**	**0.2123**	**0.0802**	**0.1754**
Δ *vs. baseline*	+23.3%	+96.3%	+39.7%	–	–	+15.4%

GRASP4Figs achieves consistent improvements across all six metrics. The BLEU-4 gain of 96.3% is notable because BLEU-4 requires exact four-gram matches, so the improvement cannot be attributed to superficial wording overlap; the METEOR gain of 39.7% corroborates this at a coarser, semantic-alignment level, indicating that OCR injection helps the model recover domain-specific vocabulary, axis quantities and legend terms, directly from the ground truth rather than paraphrasing around it. Against the strongest published configuration (BLIP-RLHF Takeaway, ROUGE-L 0.1676), our ROUGE-L of 0.1754 is still a further 4.7% improvement, confirming the gains hold against the best-tuned baseline, not only the weaker helpfulness variant.

4.2 Ablation Study

Section 3.1 argued that OCR-context injection and the composite reward address two distinct, non-overlapping deficiencies and that their combination, not either alone, should account for the aggregate gain. Table 2 tests this directly by incrementally enabling quality filtering, OCR injection, and the composite reward over the unmodified FigCaps-HF baseline.

Table 2: Ablation study on Case 1 components, full test set. ✓ denotes component enabled.

Configuration	OCR	Comp.	Q-Filt	B-4	MET	R-L
FigCaps-HF baseline				0.0190	–	0.1520
+ Quality filtering			✓	0.0175	0.1456	0.1592
+ OCR + filtering	✓		✓	0.0175	0.1456	0.1592
Ours (full)	✓	✓	✓	**0.0373**	**0.1066**	**0.1754**

Quality filtering alone (row 2) slightly *underperforms* the unfiltered baseline on BLEU-4 (0.0175 vs. 0.0190): removing 2.8% of training pairs shrinks the effective training set before either mechanism supplies the compensating signal, so this dip is expected rather than a sign that filtering is harmful. The ablation confirms the claim it was designed to test: the composite reward, not OCR injection alone, is the primary driver of the improvement, and neither mechanism reaches the full gain in isolation. Adding OCR context on top of quality filtering (row 3) yields no measurable gain over quality filtering alone, showing that merely exposing the model to additional tokens is not sufficient, it must also be pushed, via the reward signal, to use that evidence. Only once the composite reward is introduced alongside OCR injection does the full gain materialize, confirming the two-mechanism argument of Section 3.1: the combination is necessary because each mechanism addresses a deficiency the other cannot reach alone.

4.3 Case 2 Paragraph Generation Results

Having established Case 1's short-caption gains, we next evaluate whether zero-shot structured prompting, introduced in Section 3.2 as a remedy to MPGC, is actually effective at producing analytically rich paragraphs. Table 3 reports zero-shot Qwen2.5-VL-7B on all 13,355 test figures, alongside Case 1 and the collapsed LoRA fine-tuning baseline for reference.

Table 3: Case 2 paragraph generation results. ROUGE-L Recall is the primary metric. GT captions average 10–15 words; generated paragraphs average 176.1 words. *BLEU-4 is not a meaningful indicator at this length regime.

System	Len.	MET	RL-F1	RL-Rec.	B-1	B-4
Case 1 (BLIP)	≈15 w	0.1066	0.1754	–	0.2514	0.0373
Qwen+LoRA (MPGC)	≈150 w	*Coll.*	*Coll.*	*Coll.*	*Coll.*	N/A
Qwen 0-shot (Ours)	176.1 w	0.1342	0.0743	**0.3782**	0.0459	0.0034*

The ROUGE-L Recall of 0.3782 confirms that our paragraphs recover 37.8% of ground-truth caption content while additionally supplying axis descriptions, trend analysis, and conclusions absent from the short ground truth. The correspondingly low F1 of 0.0743 reflects the expected precision penalty of comparing 176-word outputs against 10–15 word references (Section 3.2), not an actual failure of content quality.

4.4 Qualitative Analysis

A recall score of 0.3782, while informative in aggregate, does not by itself show what a 37.8%-recall paragraph looks like next to the figure it describes, or whether the additional content is genuine analysis rather than noise. Figure 3 grounds this in concrete, inspectable evidence: side-by-side outputs for three representative test figures, each row showing the original figure, the author-written ground-truth caption, and the corresponding GRASP4Figs Case 2 Qwen zero-shot paragraph.

Across all three examples, the Case 2 outputs demonstrate categorically richer content: specific axis ranges, curve parameter identification, quantitative trend characterization, and explicitly articulated conclusions. The ground-truth captions, while accurate, function largely as figure titles rather than analytical explanations, precisely the gap that motivates GRASP4Figs' paragraph-generation capability.

4.5 Helpfulness Scoring Analysis

Automated n-gram metrics such as BLEU and ROUGE, while standard, measure surface overlap rather than perceived usefulness. To validate Case 1 along this complementary axis, we applied the FigCaps-HF BERT-based helpfulness scorer, the same scorer used to derive our training reward, to 20 generated captions and their ground-truth counterparts. Our generated captions scored a mean helpfulness of 2.652 on the 1–5 Likert scale versus 2.294 for the author-written ground truth, a 15.6% improvement, suggesting that OCR-augmented generation is independently judged as more helpful for figure understanding than the very captions the model was originally trained to imitate.

Taken together, these results confirm the *missing-evidence* framing from Section 3: BLIP's language decoder effectively utilizes textual tokens once supplied, even though its 384×384 encoder cannot resolve them visually, so the primary limitation is information access rather than model capacity. This suggests high-resolution encoding capable of resolving small-font text directly, rather than further scaling model capacity, is the more promising direction for reducing reliance on external OCR.

This also exposes a benchmark mismatch: FigCaps-HF targets short, 10–15 word captions, while real analytical use needs 100–200 word descriptions, and n-gram metrics penalize longer outputs by construction. The field needs evaluation methods, such as figure-grounded question answering, that do not conflate output length with output quality.

We discuss the limitations these results share, dataset domain bias, automated-scorer sample size, and MPGC's qualitative-only support, together with future work in Section 6.

5 GRASP4FIGS WEB APPLICATION

We deploy the Case 2 zero-shot pipeline as a production-ready web application for end-to-end scientific figure comprehension over research PDFs: since Case 2 requires no fine-tuning and avoids MPGC (Sections 3.1, 3.2), it is the natural choice for a system that must generalize to arbitrary uploaded PDFs rather

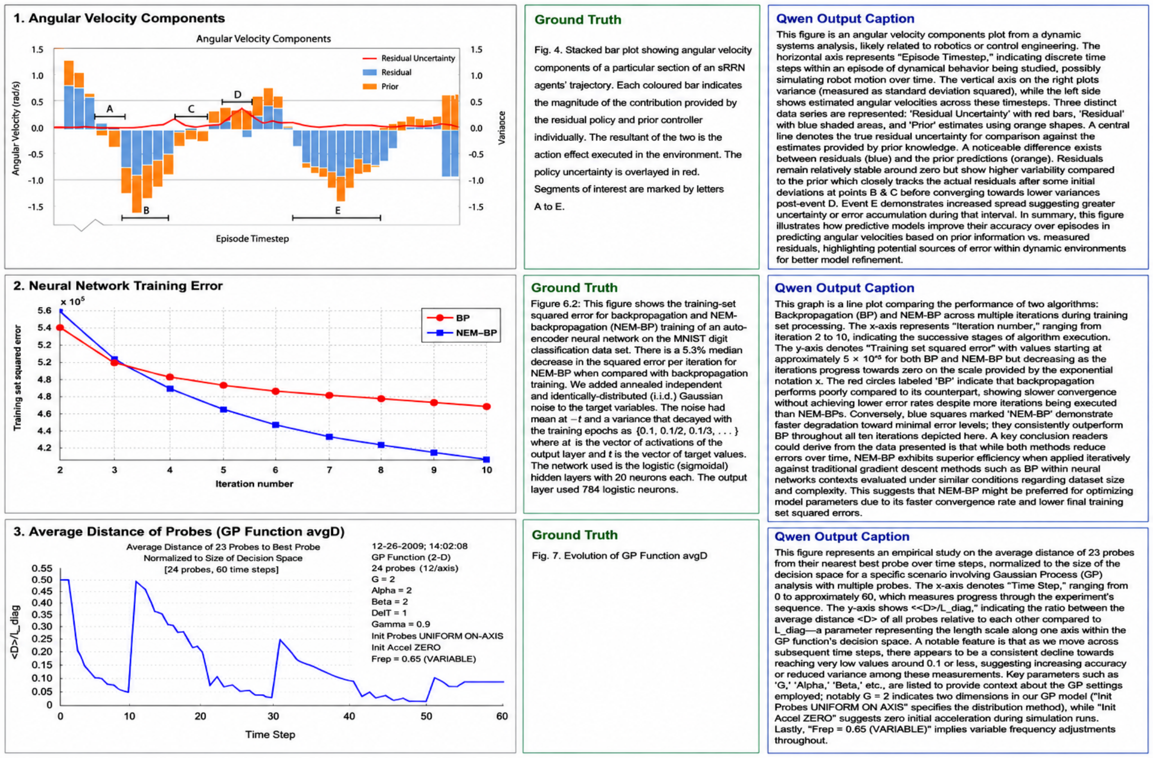

Figure 3: Qualitative results across three representative figure types. *Left*: original test figure from FigCaps-HF. *Centre*: author-written ground-truth caption (typically 10–15 words, often providing only a figure title or brief description). *Right*: GRASP4Figs Case 2 (Qwen2.5-VL-7B zero-shot) output, showing axis identification, trend characterization, and explicit scientific conclusions absent from the ground truth. Full paragraph outputs shown; average output length 176.1 words across 13,355 figures.

than a fixed benchmark. We built a full-stack pipeline around a React + Vite frontend and a FastAPI backend. Figure extraction uses PyMuPDF directly on the uploaded PDF, filtering out images below 150 px so inline icons and decorative graphics are not mistaken for scientific figures; no external OCR or annotation lookup is required here either, consistent with Case 1's inference-time independence. The backend then streams Qwen2.5-VL-7B's paragraph generation to the client via Server-Sent Events, surfacing the same token-by-token decoding used in Section 3.2 as real-time output rather than a single blocking response.

Each extracted figure includes an interactive chatbot, again powered by Qwen2.5-VL-7B, using the figure image, its generated paragraph, and conversation history as context. This extends the grounding constraint from Section 3.2: the chatbot is instructed to answer only from the image and its own prior analysis, so a multi-turn conversation does not drift into the unsupported claims mention-paragraph fine-tuning produced. A user can ask a follow-up such as "What does the crossover point indicate?" and receive a concise, grounded two-to-three sentence answer.

The application has three views: a landing page, a research dashboard surfacing the quantitative comparisons and MPGC finding from Section 4, and an analysis workspace for PDF upload, figure extraction, streaming generation, and per-figure chat. Since Qwen2.5-VL-7B is served directly rather than distilled or quantized, the analysis workspace requires an NVIDIA GPU with at least 16 GB VRAM; we have tested the full pipeline on RTX 6000 Ada hardware.

6 CONCLUSION AND FUTURE SCOPE

We presented GRASP4Figs, two case studies unified by the observation (Section 3) that short captions and long-form paragraphs fail for different reasons. Case 1's OCR-context injection and composite reward together correct FigCaps-HF's missing textual evidence and unshaped length, achieving BLEU-4 of 0.0373, a 166.4% gain over the published non-RLHF baseline and 96.3% over the strongest RLHF baseline [28, 29]. Case 2 showed that fine-tuning on SciCap+ mention-paragraphs induces MPGC, and that zero-shot structured prompting with Qwen2.5-VL [26] avoids it, achieving a ROUGE-L Recall of 0.3782. Section 5 deploys this design as a per-figure chatbot over user-uploaded PDFs.

Future work includes: generalization beyond our CS-biased test set; a large-scale, automated study of MPGC drift, shown here only qualitatively; scaling our 20-caption automated helpfulness validation; and high-resolution, OCR-free grounding [7] beyond BLIP's 384×384 encoding, with evaluation beyond n-gram overlap, e.g., figure-grounded QA.

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